

Lagrangian Formulation of the Vacuum Einstein Equations

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Summary

This note presents a self-contained derivation of the vacuum Einstein equations from the Lagrangian formulation. After reviewing the variational principle and the issue of the metric-dependent volume element, we provide detailed computations for both the Hilbert action and the Palatini action, with all intermediate algebraic steps explicitly shown.

Keywords: Lagrangian formulation, variational principle, vacuum Einstein equations, Hilbert action, Palatini action

Having discussed the Lagrangian formulations of the Klein–Gordon scalar field and the electromagnetic field[2], we now address the vacuum Einstein field equations in general relativity. The difficulty is that the action is a functional of the metric, and the volume element in the integral generally depends on the metric as well, so it cannot be treated as constant in the variation.

To resolve this issue, we may introduce a fixed volume element $e \equiv e_{abcd}$ whose components in any coordinate basis are ± 1 , i.e.,

$$e_{ijkl} = \begin{cases} 1, & ijkl \text{ is an even permutation,} \\ -1, & ijkl \text{ is an odd permutation.} \end{cases} \quad (1)$$

In other words, for any given choice of coordinates, we remove the factor $\sqrt{|g|}$ (in general relativity, $\sqrt{|g|} = \sqrt{-g}$) from the metric-adapted volume element

$$\epsilon = \sqrt{|g|} dx^1 \wedge \cdots \wedge dx^n, \quad (2)$$

thereby obtaining a volume element e that does not depend on variations of the metric.

This means that even if the metric remains unchanged, the volume element e changes whenever we adopt a new coordinate system; consequently, the "volume" of a fixed region of the manifold computed with e depends on the choice of coordinates, not on the metric. Thus it does not correspond to the intuitive notion of physical volume. In order for the action to be meaningful as an integral of a Lagrangian density, the Lagrangian density must contain a factor $\sqrt{-g}$ to compensate for the lost covariance. In the terminology of Wald's book [1], the Lagrangian density must be a *tensor density*.

An alternative, simpler way to view this procedure is to retain the adapted volume element ϵ in the integration, and merely decompose it as $\sqrt{-g} e$ in the chosen coordinate basis whenever it is convenient to do so in the variation.

The above discussion serves as a preparation of the basic concepts. In what follows, in Sec. 1, we derive the field equations by treating the action solely as a functional of the metric g^{ab} ; in Sec. 2, we treat ∇_a as an independent variable in the action and show that the same result is obtained.

1 Hilbert action

Following the first approach discussed above, namely, adopting the fixed volume element e , the Lagrangian density for the gravitational field is

$$\mathcal{L}_G = \sqrt{-g} R = \sqrt{-g} R_{ab} g^{ab}, \quad (3)$$

and the corresponding action is

$$S[g^{ab}] = \int \mathcal{L}_G e \equiv S_G. \quad (4)$$

In this formulation, only g^{ab} is regarded as the independent variable; this is the *Hilbert action*. The variation of S is

$$\delta S_G \equiv \left. \frac{dS_G}{d\lambda} \right|_{\lambda=0} = \int \left. \frac{d\mathcal{L}_G}{d\lambda} e \right|_{\lambda=0} = 0 \quad (5)$$

where

$$\boxed{\frac{d\mathcal{L}_G}{d\lambda} = \sqrt{-g}(\delta R_{ab})g^{ab} + \sqrt{-g}R_{ab}\delta g^{ab} + R\delta(\sqrt{-g})} \quad (6)$$

1.1 Computing the term $g^{ab}\delta R_{ab}$ in $\frac{d\mathcal{L}_G}{d\lambda}$

Using the Palatini identity [see Wald's Eq. (7.5.10); the proof is omitted here],

$$\delta R_{ab} = \nabla_c(\delta C^c_{ab}) - \nabla_b(\delta C^c_{ac}), \quad (7)$$

and the standard variation of the Christoffel symbol,

$$\delta C^c_{ab} = \frac{1}{2}g^{cd}(\nabla_a\delta g_{bd} + \nabla_b\delta g_{ad} - \nabla_d\delta g_{ab}), \quad (8)$$

we now perform the contraction $g^{ab}\delta R_{ab}$ in detail.

Contracting (7) with g^{ab} , we have

$$\begin{aligned} g^{ab}\delta R_{ab} &= g^{ab}\nabla_c(\delta C^c_{ab}) - g^{ab}\nabla_b(\delta C^c_{ac}) \\ &= \nabla_c(g^{ab}\delta C^c_{ab}) - \nabla_b(g^{ab}\delta C^c_{ac}) \end{aligned}$$

$$\begin{aligned}
&= \nabla_c \left(g^{ab} \delta C^c_{ab} - g^{ac} \delta C^b_{ab} \right) \\
&\equiv \nabla_c v^c
\end{aligned} \tag{9}$$

Where

$$v^c \equiv g^{ab} \delta C^c_{ab} - g^{ac} \delta C^b_{ab}. \tag{10}$$

Now substitute (8) into (10), the first term

$$\begin{aligned}
g^{ab} \delta C^c_{ab} &= \frac{1}{2} g^{ab} g^{cd} (\nabla_a \delta g_{bd} + \nabla_b \delta g_{ad} - \nabla_d \delta g_{ab}) \\
&= \frac{1}{2} g^{cd} \left(g^{ab} \nabla_a \delta g_{bd} + g^{ab} \nabla_b \delta g_{ad} - g^{ab} \nabla_d \delta g_{ab} \right).
\end{aligned} \tag{11}$$

Using index symmetry, the first two terms inside the parentheses are equal, so

$$g^{ab} \delta C^c_{ab} = g^{cd} g^{ab} \nabla_a \delta g_{bd} - \frac{1}{2} g^{cd} \nabla_d (\delta g), \tag{12}$$

where $\delta g \equiv g^{ab} \delta g_{ab}$.

For the second term in (10), we have

$$\begin{aligned}
g^{ac} \delta C^b_{ab} &= \frac{1}{2} g^{ac} g^{bd} (\nabla_a \delta g_{bd} + \nabla_b \delta g_{ad} - \nabla_d \delta g_{ab}) \\
&= \frac{1}{2} g^{ac} g^{bd} \nabla_a \delta g_{bd} + \frac{1}{2} g^{ac} g^{bd} \nabla_b \delta g_{ad} - \frac{1}{2} g^{ac} g^{bd} \nabla_d \delta g_{ab}.
\end{aligned} \tag{13}$$

Now relabel indices in the second and third terms of (13). Since in the second term, a and b are dummy indices, we may swap $a \leftrightarrow b$, obtaining

$$\frac{1}{2} g^{ac} g^{bd} \nabla_b \delta g_{ad} = \frac{1}{2} g^{bc} g^{ad} \nabla_a \delta g_{bd}. \tag{14}$$

In the third term, similarly, we may swap $a \leftrightarrow d$ and $b \leftrightarrow a$, yielding

$$-\frac{1}{2} g^{ac} g^{bd} \nabla_d \delta g_{ab} = -\frac{1}{2} g^{dc} g^{ba} \nabla_a \delta g_{ab} = -\frac{1}{2} g^{cd} \nabla_d (\delta g). \tag{15}$$

Thus

$$\begin{aligned}
g^{ac} \delta C^b_{ab} &= \frac{1}{2} g^{ac} g^{bd} \nabla_a \delta g_{bd} + \frac{1}{2} g^{bc} g^{ad} \nabla_a \delta g_{bd} - \frac{1}{2} g^{cd} \nabla_d (\delta g) \\
&= g^{ac} g^{bd} \nabla_a \delta g_{bd} - \frac{1}{2} g^{cd} \nabla_d (\delta g).
\end{aligned} \tag{16}$$

Subtract (16) from (12), we obtain

$$\begin{aligned}
v^c &= \left[g^{cd} g^{ab} \nabla_a \delta g_{bd} - \frac{1}{2} g^{cd} \nabla_d (\delta g) \right] - \left[g^{ac} g^{bd} \nabla_a \delta g_{bd} - \frac{1}{2} g^{cd} \nabla_d (\delta g) \right] \\
&= g^{cd} g^{ab} \nabla_a \delta g_{bd} - g^{ac} g^{bd} \nabla_a \delta g_{bd}.
\end{aligned} \tag{17}$$

In the second term of (17), relabel $a \leftrightarrow c$, obtaining

$$g^{ac} g^{bd} \nabla_a \delta g_{bd} = g^{cd} g^{ba} \nabla_c \delta g_{ba} = g^{cd} g^{ab} \nabla_c \delta g_{ab}. \quad (18)$$

Therefore

$$v^c = g^{cd} g^{ab} (\nabla_a \delta g_{bd} - \nabla_c \delta g_{ab}). \quad (19)$$

Lowering the index c and relabeling, we obtain

$$v_a = \nabla^b (\delta g_{ab}) - g^{cd} \nabla_a (\delta g_{cd}). \quad (20)$$

Summarizing the results of (9) and (20), we obtain

$$\boxed{g^{ab} \delta R_{ab} = \nabla^a v_a, \quad v_a = \nabla^b (\delta g_{ab}) - g^{cd} \nabla_a (\delta g_{cd}).} \quad (21)$$

1.2 Computing the term $R\delta(\sqrt{-g})$ in $\frac{d\mathcal{L}_G}{d\lambda}$

Let g denote the determinant of the metric matrix $g_{\mu\nu}$. Using the standard formula for the derivative of a determinant,

$$\frac{\partial g}{\partial g_{\mu\nu}} = g g^{\mu\nu}, \quad (22)$$

we have

$$\delta g = g g^{\mu\nu} \delta g_{\mu\nu}. \quad (23)$$

Since $\sqrt{-g} = (-g)^{1/2}$, its variation is

$$\begin{aligned} \delta(\sqrt{-g}) &= \frac{1}{2\sqrt{-g}} \delta(-g) \\ &= -\frac{1}{2\sqrt{-g}} \delta g. \end{aligned} \quad (24)$$

Substituting (23) into (24) gives

$$\delta(\sqrt{-g}) = -\frac{1}{2\sqrt{-g}} g g^{\mu\nu} \delta g_{\mu\nu} = \frac{1}{2} \sqrt{-g} g^{\mu\nu} \delta g_{\mu\nu}. \quad (25)$$

Renaming $(\mu, \nu) \rightarrow (a, b)$, we obtain

$$\delta(\sqrt{-g}) = \frac{1}{2} \sqrt{-g} g^{ab} \delta g_{ab}. \quad (26)$$

From $g^{ab} g_{bc} = \delta^a_c$, taking the variation gives

$$(\delta g^{ab}) g_{bc} + g^{ab} (\delta g_{bc}) = 0. \quad (27)$$

Multiplying (27) by g^{cd} and contracting yields

$$\delta g^{ab} = -g^{ac} g^{bd} \delta g_{cd}, \quad (28)$$

and conversely,

$$\delta g_{ab} = -g_{ac}g_{bd}\delta g^{cd}. \quad (29)$$

Substituting (29) into (26), we get

$$\begin{aligned} \delta(\sqrt{-g}) &= \frac{1}{2}\sqrt{-g} g^{ab} \left(-g_{ac}g_{bd}\delta g^{cd}\right) \\ &= -\frac{1}{2}\sqrt{-g} g_{cd}\delta g^{cd}. \end{aligned} \quad (30)$$

Renaming $(c, d) \rightarrow (a, b)$, we obtain that

$$\delta(\sqrt{-g}) = -\frac{1}{2}\sqrt{-g} g_{ab}\delta g^{ab}. \quad (31)$$

Summarizing the results of (26) and (31), we obtain

$$\boxed{\delta(\sqrt{-g}) = \frac{1}{2}\sqrt{-g} g^{ab}\delta g_{ab} = -\frac{1}{2}\sqrt{-g} g_{ab}\delta g^{ab}.} \quad (32)$$

1.3 Combining the Previous Calculations

Substituting (21) and (32) into (6), and then substituting the result into (5), we obtain

$$\frac{dS_G}{d\lambda} = \int \frac{d\mathcal{L}_G}{d\lambda} e = \int \nabla^a v_a \sqrt{-g} e + \int \left(R_{ab} - \frac{1}{2}Rg_{ab}\right) \delta g^{ab} \sqrt{-g} e \quad (33)$$

The first term in the above equation is the integral of the divergence $\nabla^a v_a$ with respect to the adapted volume element $\epsilon = \sqrt{-g} e$. By Stokes' theorem, we have

$$\int_U \nabla_a v^a \epsilon = \int_{\dot{U}} v_a n^a, \quad (34)$$

where n^a is the unit normal to the boundary $\dot{U}$ (assumed to be non-null). Using (21), on $\dot{U}$ we have

$$\begin{aligned} v_a n^a &= n^a g^{bc} [\nabla_c(\delta g_{ab}) - \nabla_a(\delta g_{bc})] \\ &= n^a (h^{bc} + n^b n^c) [\nabla_c(\delta g_{ab}) - \nabla_a(\delta g_{bc})] \\ &= n^a h^{bc} [\nabla_c(\delta g_{ab}) - \nabla_a(\delta g_{bc})] \\ &= -n^a h^{bc} \nabla_a(\delta g_{bc}), \end{aligned} \quad (35)$$

where $h_{ab} = g_{ab} \pm n_a n_b$ is the induced metric on $\dot{U}$. In the third step above we used

$$\begin{aligned} n^a n^b n^c [\nabla_c(\delta g_{ab}) - \nabla_a(\delta g_{bc})] &= n^a n^b n^c [\nabla_c(\delta g_{ab}) - \nabla_a(\delta g_{cb})] \\ &= 2n^c n^a n^b \nabla_{[c} \delta g_{a]b} = 0 \end{aligned} \quad (36)$$

and the fourth step used the boundary condition $\delta g_{ab} = 0$ on $\dot{U}$. Let

$$K \equiv K^a_a = h^a_b \nabla_a n^b. \quad (37)$$

(This is the trace of the extrinsic curvature of the hypersurface.) Then

$$\begin{aligned}
 \delta K &= h^a{}_b (\delta C)^b{}_{ac} n^c \\
 &= \frac{1}{2} n^c h^a{}_b g^{bd} [\nabla_a (\delta g_{cd}) + \nabla_c (\delta g_{ad}) - \nabla_d (\delta g_{ac})] \\
 &= \frac{1}{2} n^c h^{ad} \nabla_c (\delta g_{ad})
 \end{aligned} \tag{38}$$

In the third equality of (38), the first and third terms vanish on the boundary because they contain the factors $h^{ad} \nabla_a (\delta g_{cd})$ and $h^{ad} \nabla_d (\delta g_{ac})$, respectively, which are precisely the tangential covariant derivatives of δg_{ab} along the boundary; since $\delta g_{ab} = 0$ on the entire boundary $\dot{U}$, its tangential derivatives vanish. Therefore, for metric variations satisfying $\delta g_{ab} = 0$ on $\dot{U}$, from (33), (35), and (38) we obtain

$$\frac{dS_G}{d\lambda} = -2 \int_{\dot{U}} \delta K + \int_U G_{ab} \delta g^{ab} \epsilon \tag{39}$$

Notice that there is a term $-2 \int_{\dot{U}} \delta K$ in the above equation. To eliminate this term, it suffices to redefine the action as

$$S'_G = S_G + 2 \int_{\dot{U}} K \tag{40}$$

Under the new action, the variation becomes

$$\frac{dS'_G}{d\lambda} = \int_U G_{ab} \delta g^{ab} \epsilon \tag{41}$$

Setting this expression to zero yields $G_{ab} = 0$, which is the vacuum Einstein equation.

2 Palatini action

As shown in §1.3, if we stipulate that the derivative operator is compatible with the metric (i.e., treating ∇_a as a function of g^{ab} and thereby eliminating its independence), then when the Lagrangian density is regarded as a functional of g^{ab} alone, a modification of S_G is required to derive the correct Einstein field equations.

If, on the other hand, we do not impose $\nabla_a g_{bc} = 0$ prior to the calculation—that is, we do not require the derivative operator to be metric-compatible, but instead treat it as a second independent variable—then R_{ab} can be viewed as a functional of ∇_a alone, since

$$R_{ac} = -2\nabla_{[a} C^b{}_{b]c} + 2C^e{}_{c[a} C^b{}_{b]e}, \tag{42}$$

[see Wald's Eq. (7.5.9)] is a geometric quantity determined by the connection, not by the metric.

Keeping the same expression for the Lagrangian density, we may rewrite it as

$$\mathcal{S}_G[g^{ab}, \nabla_a] = \int \sqrt{-g} R_{ab} g^{ab} \epsilon, \tag{43}$$

and the corresponding S_G is called the *Palatini action*.

In this section, we will find that by introducing the two independent variables as described above, there is no need to modify S_G in order to obtain the Einstein equations naturally. At $\lambda = 0$, we obtain

$$\begin{aligned}
\frac{d\mathcal{S}_G}{d\lambda} &= \int [\sqrt{-g}(\delta R_{ab})g^{ab} + \sqrt{-g}R_{ab}\delta g^{ab} + R\delta(\sqrt{-g})] e \\
&= \int \left[\sqrt{-g}(\delta R_{ab})g^{ab} + \left(R_{ab} - \frac{1}{2}Rg_{ab} \right) \delta g^{ab} \sqrt{-g} \right] e \\
&= \int \sqrt{-g}(\delta R_{ab})g^{ab} e + \int \left(R_{ab} - \frac{1}{2}Rg_{ab} \right) \delta g^{ab} \sqrt{-g} e \\
&= -2 \int g^{ab} \nabla_{[a} \delta C^c_{c]b} \sqrt{-g} e + \int \left(R_{ab} - \frac{1}{2}Rg_{ab} \right) \delta g^{ab} \sqrt{-g} e \\
&= -2 \int g^{ab} \tilde{\nabla}_{[a} \delta C^c_{c]b} \sqrt{-g} e \\
&\quad + \int g^{ab} [C^d_{ab} \delta C^c_{cd} + C^c_{cd} \delta C^d_{ab} - 2C^d_{cb} \delta C^c_{ad}] \sqrt{-g} e \\
&\quad + \int \left(R_{ab} - \frac{1}{2}Rg_{ab} \right) \delta g^{ab} \sqrt{-g} e
\end{aligned} \tag{44}$$

where the fourth equality uses Wald's Eq. (7.5.10), which is Eq. (7) in this note. The fifth equality uses

$$\nabla_a T^{b_1 \dots b_k}_{c_1 \dots c_l} = \tilde{\nabla}_a T^{b_1 \dots b_k}_{c_1 \dots c_l} + \sum_i C^{b_i}_{ad} T^{b_1 \dots d \dots b_k}_{c_1 \dots c_l} - \sum_j C^d_{ac_j} T^{b_1 \dots b_k}_{c_1 \dots d \dots c_l}. \tag{45}$$

If we require that $\tilde{\nabla}_a$ be compatible with the metric, then we have

$$\begin{aligned}
\frac{d\mathcal{S}_G}{d\lambda} &= -2 \int \tilde{\nabla}_{[a} g^{ab} \delta C^c_{c]b} \sqrt{-g} e \\
&\quad + \int g^{ab} [C^d_{ab} \delta C^c_{cd} + C^c_{cd} \delta C^d_{ab} - 2C^d_{cb} \delta C^c_{ad}] \sqrt{-g} e \\
&\quad + \int \left(R_{ab} - \frac{1}{2}Rg_{ab} \right) \delta g^{ab} \sqrt{-g} e \\
&= -2 \int \tilde{\nabla}_{[a} g^{(ab)} \delta C^c_{b]c} \sqrt{-g} e \\
&\quad + \int g^{ab} [C^d_{ab} \delta C^c_{cd} + C^c_{cd} \delta C^d_{ab} - 2C^d_{cb} \delta C^c_{ad}] \sqrt{-g} e \\
&\quad + \int \left(R_{ab} - \frac{1}{2}Rg_{ab} \right) \delta g^{ab} \sqrt{-g} e \\
&= \int g^{ab} [C^d_{ab} \delta C^c_{cd} + C^c_{cd} \delta C^d_{ab} - 2C^d_{cb} \delta C^c_{ad}] \sqrt{-g} e \\
&\quad + \int \left(R_{ab} - \frac{1}{2}Rg_{ab} \right) \delta g^{ab} \sqrt{-g} e \\
&= \int [C^{bd}_{d} \delta^a_c + C^d_{dc} g^{ab} - 2C^b_c{}^a] \delta C^c_{ab} \sqrt{-g} e
\end{aligned}$$

$$+ \int \left(R_{ab} - \frac{1}{2} R g_{ab} \right) \delta g^{ab} \sqrt{-g} e. \quad (46)$$

Here the first equality uses the compatibility of g^{ab} with $\tilde{\nabla}_a$; the second equality uses the symmetry of g^{ab} and the symmetry of C^c_{ab} in its two lower indices; the third equality uses the fact that the contraction of a symmetric and an antisymmetric pair of indices vanishes. The detailed calculation for the fourth equality is as follows. The integrand on the third line is

$$g^{ab} \left[C^d_{ab} \delta C^c_{cd} + C^c_{cd} \delta C^d_{ab} - 2 C^d_{cb} \delta C^c_{ad} \right]. \quad (47)$$

The first term can be written as

$$g^{ab} C^d_{ab} \delta C^c_{cd} = g^{ab} C^d_{ab} \delta C^c_{cd} \xrightarrow{d \rightarrow b} g^{ab} C^b_{ab} \delta C^c_{cb} = (C^{bd}_d \delta^a_c) \delta C^c_{ab}. \quad (48)$$

Thus it contributes the coefficient $C^{bd}_d \delta^a_c$. The second term can be written as

$$g^{ab} C^c_{cd} \delta C^d_{ab} \xrightarrow{d \leftrightarrow c} g^{ab} C^d_{dc} \delta C^c_{ab} = (C^d_{dc} g^{ab}) \delta C^c_{ab}. \quad (49)$$

Thus it contributes the coefficient $C^d_{dc} g^{ab}$. The third term can be written as

$$\begin{aligned} -2 g^{ab} C^d_{cb} \delta C^c_{ad} &\xrightarrow{d \leftrightarrow b} -2 g^{ad} C^b_{cb} \delta C^c_{ad} \\ &\xrightarrow{d \rightarrow b} -2 g^{ab} C^b_{cb} \delta C^c_{ab} \\ &= (-2 C^b_c{}^a) \delta C^c_{ab}. \end{aligned} \quad (50)$$

Thus it contributes the coefficient $-2 C^b_c{}^a$. Adding the three coefficients together, we obtain

$$\boxed{C^{bd}_d \delta^a_c + C^d_{dc} g^{ab} - 2 C^b_c{}^a} \quad (51)$$

multiplied by δC^c_{ab} , which is precisely the integrand on the fourth line.

Since δC^c_{ab} is symmetric in a and b , the contribution from the antisymmetric part with respect to a, b vanishes automatically. Therefore the variation with respect to δC^c_{ab} requires the symmetric part of the bracket in (46) to vanish, namely

$$C^{(b|d|}{}_d \delta^a)_{c} + C^d_{dc} g^{(ab)} - 2 C^{(b}_c{}^a) = 0. \quad (52)$$

or equivalently,

$$C^{bd}_d \delta^a_c + C^{ad}_d \delta^b_c + C^d_{dc} g^{ab} + C^d_{dc} g^{ba} - C^b_c{}^a - C^a_c{}^b = 0. \quad (53)$$

Using $C^d_{dc} = C^b_{bc}$ (by relabeling) and $C^{ad}_d = C^a_{bd} g^{bd} = C^a_b{}^b$ (by index relabeling), this simplifies to

$$C^{bd}_d \delta^a_c + C^{ad}_d \delta^b_c + 2 C^d_{dc} g^{ab} - C^b_c{}^a - C^a_c{}^b = 0. \quad (54)$$

Contracting the symmetrized equation with g_{ab} , we have

$$g_{ab} \left(C^{bd}_d \delta^a_c + C^{ad}_d \delta^b_c + 2 C^d_{dc} g^{ab} - C^b_c{}^a - C^a_c{}^b \right)$$

$$\begin{aligned}
&= 2C_c^d + 8C_{dc}^d - C_{cb}^b - C_{ca}^a \\
&= 2g^{ce}g_{df}C_{df}^e + 6\delta_e^d\delta_c^f C_{df}^e \\
&= (2g^{ce}g_{df} + 6\delta_e^d\delta_c^f)C_{df}^e = 0.
\end{aligned} \tag{55}$$

and hence

$$C_{ab}^c = 0. \tag{56}$$

Therefore $\nabla_a = \tilde{\nabla}_a$. We have already stipulated that $\tilde{\nabla}_a$ is the derivative operator compatible with the metric. Therefore, $\delta\mathcal{S}_G/\delta C_{ab}^c = 0$ naturally implies that ∇_a is compatible with the metric.

As before, $\delta\mathcal{S}_G/\delta g^{ab} = 0$ yields the Einstein equation.

References

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A Appendix

A.1 Report on Use of AI Tools

I primarily use AI for simple repetitive calculations, as well as for English translation and polishing. AI also assists me in explaining concepts, verifying my ideas, and providing valuable proof strategies.

A.2 What Is New in This Work

This text is my reading notes on Wald's book [1], intended to supplement the specific computational details, while following the basic content of the original work.

Specifically, prior to Sec. 1, I have rephrased Wald's discussion on the volume element being independent of the metric *entirely in my own words*.

In Sec. 1, I begin by presenting a detailed calculation of Wald's Eqs. (E.1.15) and (E.1.16) in §1.1, starting from Wald's Eq. (7.5.10) and standard identities of differential geometry. I also provide a proof of Eq. (E.1.17) in §1.2. Section 1.3 mainly follows the content of Wald's book, with only minor modifications in wording and some supplementary details for straightforward computations [e.g., Eq. (36)].

In Sec. 2, I explain, in my own words, the rationale for introducing the two independent variables and why R_{ac} can be regarded as a functional of ∇_a alone. I also supplement many computational details leading to Eq. (46) [from Eq. (44) to Eq. (51)]. In addition, I provide a detailed derivation showing why $\delta\mathcal{S}_G/\delta C_{ab}^c = 0$ implies $\nabla_a = \tilde{\nabla}_a$.

Finally, it should be emphasized that this note is merely a set of reading notes on Wald's book; the overall logic and ideas are inherited from Wald's work. My contribution consists only of rephrasing, reorganizing, and supplementing the specific computational and proof details. If there is anything of value in this note, the credit should first go to Wald's book. Should the copyright holder believe that this note infringes upon the copyright of Wald's book, please contact me and I will remove it promptly.